PRACTICE SHEET

1. A vertical pole with height more than 100 consists of two parts, the lower being one – third of the whole. At a point on a horizontal plane through the foot and 40m from it, the upper part subtends an angle whose tangent is1/2. What is the height of the pole?
(a) 110m (b) 200m
(c) 120 m (d) 150m

2. The angle of elevation of the top of a pillar of height h at a point on the ground at a distance x from the pillar is 30°. On walking a distance 'd' towards the pillar the angle of elevation becomes 60°. Then, which one of the following is correct?
(a) $x = d + h$ (b) $x = \frac{3d}{2}$
(c) $x = \frac{5d}{4}$ (d) $x = 2d$

3. The angle of elevation of the top of a tower EF (F being the foot of the tower) as seen from a point A which is on the same level as F, is α. On advancing towards the foot of the tower the angle of elevation of the top of the tower as seen from a point B such that AB = x, is β. If BF = y, h is the height of the tower and $\alpha + \beta = \pi/2$, then which one of the following is correct?
(a) $h^2 = x^2 + xy$ (b) $h = y^2 + xy^2$
(c) h2 = y2 + xy (d) h = y + x2y

4. The lower 24 m portion of a 50m tall tower is painted green and the remaining portion red. What is the distance of a point on the ground from the base of the tower where the two different portions of the tower subtend equal angles?
(a) 60m (b) 72m
(c) 90m (d) 120m

5. What should be the height of a flag where a 20 feet long ladder reaches 20 feet below the flag (The angle of elevation of the top of the flag at the foot of the ladder is 60°)?
(a) 20 feet (b) 30 feet
(c) 40 feet (d) $20\sqrt{2}$ feet

6. PT, a tower of height 2^x meter, P being the foot, T being the top of the tower. A, B are points on the same line with P. If AP = 2^{x+1}m, BP = 192 m and if the angle of elevation of the tower as seen from b is double the angle of the elevation of the tower as seen from A, then what is the value of x?
(a) 6 (b) 7
(c) 8 (d) 9

7. The foot of a tower of height *h* m is in a direct line between two observers A and B. If the angles of elevation of the top of the tower as seen from A and B are α and β respectively and if AB = *d* m, then what is h / d equal to?
(a) $\frac{\tan(\alpha+\beta)}{(\cot\alpha\cot\beta-1)}$ (b) $\frac{\cot(\alpha+\beta)}{(\cot\alpha\cot\beta-1)}$
(c) $\frac{\tan(\alpha+\beta)}{(\cot\alpha\cot\beta+1)}$ (d) $\frac{\cot(\alpha+\beta)}{(\cot\alpha\cot\beta+1)}$

8. A man observes the elevation of a balloon to be 30°. He then walks 1 km towards the balloon and finds that the elevation is 60°. What is the height of the balloon?
(a) 1/2 km (b) $\sqrt{3}$ /2 km
(c) 1/3 km (d) 1 km

9. The angle of elevation from a point on the bank of a river of the top of a temple on the other bank is 45°. Retreating 50m, the observer finds the new angle of elevation as 30°. What is the width of the river?
(a) 50m (b) $50\sqrt{3}$m
(c) $50/(\sqrt{3}-1)$m (d) 100m

10. Looking from the top of a 20 m high building, the angle of elevation of the top of a tower is 60° and the angle of depression of its bottom is 30°. What is he height of the tower?
(a) 50m (b) 60m
(c) 70m (d) 80m

11. Two houses are collinear with the base of a tower and are at distance 3m and 12m from the base of the tower. The angles of elevation from these two houses of the top of the tower are complementary. What is the height of the tower?
(a) 4m (b) 6m
(c) 7.5m (d) 36m

12. The angle of depression of vertex of a regular hexagon lying in a horizontal plane, from the top of tower of height 75m located at the centre of the regular hexagon is 60°. What is the length of each side of the hexagon?
(a) $50\sqrt{3}$m (b) 75m
(c) $25\sqrt{3}$m (d) 25 m

13. A ladder of 17ft length reaches a window which is 15ft above the ground on one side of the street. Keeping its foot at the same point the ladder is turned to the other side of the street and now it reaches a window 8 ft high. What is the width of the street?
(a) 23ft (b) 15ft
(c) 25ft (d) 30ft

14. The heights of two trees are x and y, where x > y. The tops of the trees are at a distance *z* apart. If s is the shortest distance between the trees, then what is s^2 equal to?
(a) $x^2 - y^2 - z^2 - 2xy$ (b) $x^2 + y^2 - z^2$
(c) $x^2 + y^2 + z^2 - 2xy$ (d) $z^2 - x^2 - y^2 + 2xy$

15. From a certain point on a straight road, a persons observe a tower in the West direction at a distance of 200 m. He walks some distance along the road and finds that the same tower is 300 m South of him. What is the shortest distance of the tower from the road?
(a) $\frac{300}{\sqrt{13}}$m (b) $\frac{500}{\sqrt{13}}$m
(c) $\frac{600}{\sqrt{13}}$m (d) $\frac{900}{\sqrt{13}}$m

16. Two poles are 10 m and 20 m high. The line joining their tops makes an angle of 15^0 with the horizontal. What is the approximate distance between the poles?

(a) 35.3m (b) 37.3 m
(c) 41 m (d) 44 m

17. From the top of a lighthouse 120 m above the sea, the ae of depression of a boat is 15^0. What is the distance of the boat from the lighthouse?

(a) 400 m (b) 421 m
(c) 444 m (d) 460 m

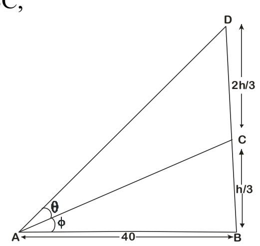
18. The angle of elevation of the top of a flag post from a point 5 m away from its base is 75^0. What is the approximate height of the flag post?

(a) 15 m (b) 17 m
(c) 19 m (d) 21 m

ANSWER KEYS																			
1.	c	**2.**	b	**3.**	c	**4.**	d	**5.**	b	**6.**	c	**7.**	b	**8.**	b	**9.**	c	**10.**	d
11.	b	**12.**	c	**13.**	a	**14.**	d	**15.**	c	**16.**	b	**17.**	c	**18.**	c				

Solutions

Sol. 1. (c)
Let h be the height of pole, upper portion CD subtend angle θ at A.
Then, $\tan\theta = 1/2$
Let lower part BC subtend angle ϕ at A then
In Δ ABC,



$$\tan\phi = \frac{BC}{AB} = \frac{h/3}{40} = \frac{h}{120}$$

In ΔABD, $\tan(\theta+\phi) = \frac{BD}{AB}$

$$\Rightarrow \frac{\tan\theta + \tan\phi}{1-\tan\theta\tan\phi} = \frac{h}{40}$$

$$\Rightarrow \frac{\frac{1}{2}+\frac{h}{120}}{1-\frac{h}{240}} = \frac{h}{40}$$

$$\Rightarrow \frac{2(60+h)}{(240-h)} = \frac{h}{40}$$

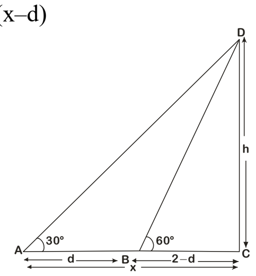
$\Rightarrow 80(60 + h) = 240h - h^2 \Rightarrow 4800 + 80h = 240h - h^2$
$\Rightarrow h^2 - 160h + 4800 = 0 \Rightarrow (h-120)(h-40) = 0$
$\Rightarrow h = 120$
[h = 40 is discarded, since h > 100 is given]

Sol. 2. (b)
Let DC be the pillar of height h and A be the point a distance x from pillar such that ∠CAD = 30°. On walking a distance d towards pillar (point B) ∠ CBD = 60°. So, in ΔBCD,

$$\tan 60° = \frac{CD}{BC}$$

$$\Rightarrow \sqrt{3} = \frac{h}{x-d}$$

$\Rightarrow h = \sqrt{3}(x-d)$



and in Δ ACD,

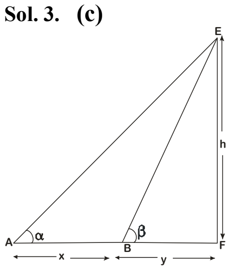
$$\tan 30° = \frac{CD}{AC}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{x}$$

$\Rightarrow x = h\sqrt{3}$
$\Rightarrow x = 3(x-d)$ (using Eq. (i))
$\Rightarrow x = 3x - 3d$
$\Rightarrow 2x = 3d$

$$\Rightarrow x = \frac{3d}{2}$$

Sol. 3. (c)



Let EF be the height of the tower, and FB = Y, AB = x and EF = h.
In ΔBEF,

$$\tan\beta = \frac{EF}{BF}$$

$\tan\beta = h/y$ …(i)

and in Δ AFE,

$$\tan\alpha = \frac{EF}{AF}$$

$$\Rightarrow \tan\alpha = \frac{h}{x+y}$$

$\Rightarrow \tan\alpha\left(\frac{\pi}{2}-\beta\right) = \frac{h}{x+y}$ (given than $\alpha+\beta=\pi/2$)

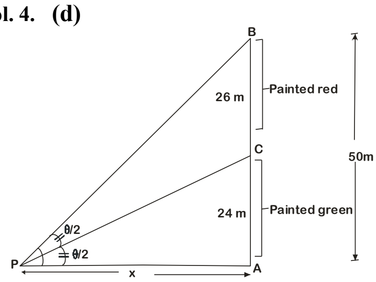
$\Rightarrow \cot\beta = \frac{h}{x+y}$ …(ii)

From Eqs. (i) and (ii) we get

$$\tan\beta \cdot \cot\beta = \frac{h}{y}\cdot\frac{h}{(x+y)}$$

($\because \tan\beta \cdot \cot\beta = 1$)
$\Rightarrow xy + y^2 = h^2$

Sol. 4. (d)



Let the distance, be x, and angle APB = θ, then ∠BPC = ∠ APC = θ/2
In triangle ΔAPB

$\tan\theta = \frac{AB}{x} = \frac{50}{x}$ …(i)

and in triangle APC

$\tan\frac{\theta}{2} = \frac{AC}{x} = \frac{24}{x}$ …(ii)

$\tan\theta = \frac{2\tan\frac{\theta}{2}}{1-\tan^2\frac{\theta}{2}}$ …(iii)

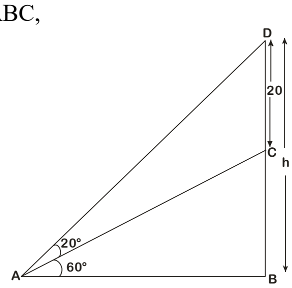
Putting the value of tan θ and tan $\frac{\theta}{2}$
From equation (1) and (2) in equation (3)

$$\frac{50}{x} = \frac{2\times\frac{24}{x}}{1-\left(\frac{24}{x}\right)^2}$$

$$\text{Or, } \frac{50}{x} = \frac{48}{x}\times\frac{x^2}{x^2-(24)^2}$$

or, $50\{x^2 - (24)^2\} = 48x^2$
or $50x^2 - 50 \times (24)^2 = 48x^2$
or, $2x^2 = (24)^2 \times 50$
$x = 25 \times (24)^2$
$x = |5\times24| = 120$ m
Such a point can exist on the either side of the tower.

Sol. 5. (b)
In Δ ABC,



$$\tan 60° = \frac{BD}{AD}$$

$$\Rightarrow \sqrt{3} = \frac{h}{AB}$$

$$\Rightarrow AB = \frac{h}{\sqrt{3}}$$

$$\Rightarrow AB = \frac{h}{3}\sqrt{3}$$

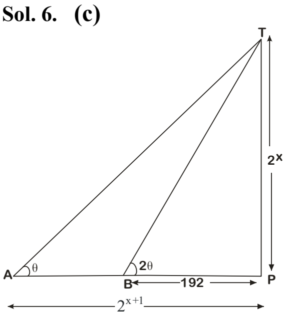
Now, in ΔABC, $AC^2 = AB^2 + BC^2$

$$\Rightarrow 20^2 = \left(\frac{h}{\sqrt{3}}\right)^2 + (h-20)^2$$

$\Rightarrow h^2 + 3h^2 - 120h = 0$
$\Rightarrow 4h^2 - 120h = 0$
$\Rightarrow h(h-30) = 0$
h = 0 or 30
h = 0 or 30

h = 0 or 30
h = 0 not possible
$\Rightarrow$ h = 30 ft

Sol. 6. (c)



Let PTbe the tower such that PT=2^xm, AP=2^{x+1} m
A and B are two points joining foot of the tower.
T is top of tower.

Also, given BP = 192 m, $\angle$TAP = θ and $\angle$TBP = 2θ

$\tan\theta = \frac{PT}{AP} \Rightarrow \tan\theta = \frac{2^x}{2^{x+1}} = \frac{1}{2}$

Now, in ΔPTB

$\tan 2\theta = \frac{PT}{AP} \Rightarrow \tan\theta = \frac{2^x}{2^{x+1}} = \frac{1}{2}$

Now, in ΔPTB

$\tan 2\theta = \frac{PT}{PB} = \frac{2^x}{192}$

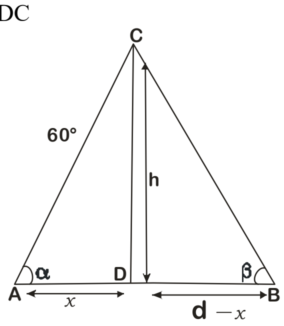
$\Rightarrow \frac{2(1/2)}{1-\frac{1}{4}} = \frac{2^x}{192} \Rightarrow \frac{1}{3/4} = \frac{2x}{192} = \frac{4}{3}\times 192 = 2^x$

$2^x = 256$

$\Rightarrow 2^x = 2^8 \Rightarrow x = 8$

Sol. 7. (b)

Let AD = x
$\therefore$ DB = d – x
In Δ ADC



$\tan\alpha = \frac{h}{x} \quad \Rightarrow x = h\cot\alpha$

In Δ CDB, $\tan\beta = \frac{h}{d-x}$

$\Rightarrow$ d – x = h cot β
From equation (i) and (ii), we get
d = h (cot α + cot β)
We know,

$\cot(\alpha+\beta) = \frac{\cot\alpha + \cot\beta - 1}{\cot\beta + \cot\alpha}$

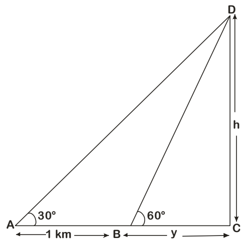
$\Rightarrow \cot\beta = \cot\alpha = \frac{\cot\alpha + \cot\beta - 1}{\cot(c)}$

$\therefore d = h\left[\frac{\cot\alpha + \cot\beta - 1}{\cot(\alpha+\beta)}\right]$

$\Rightarrow \frac{h}{d} = \frac{\cot(\alpha+\beta)}{\cot\alpha + \cot\beta - 1}$

Sol. 8. (b)

Let the height of the balloon be h and new distance between BC by as shown in figure given below,



$\tan 60° = \frac{CD}{BC}$

$\Rightarrow \sqrt{3} = h/y \Rightarrow h = \sqrt{3}y$

And now in ΔADC, $\tan 30° = \frac{CD}{AC}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{1+y}$

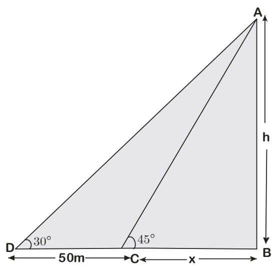
$\Rightarrow 1+y = h\sqrt{3} \Rightarrow 1 + y = 3y$

$\Rightarrow y = ½$

$\therefore h = \frac{\sqrt{3}}{2}$

Sol. 9. (c)

In ΔABC, $\tan 45° = \frac{AB}{BC}$



$\Rightarrow 1 + \frac{h}{x}$

$\Rightarrow$ h = x …(i)

Now, in ΔABD,

$\tan 30° = \frac{AB}{BD}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{x+50}$

$\Rightarrow x + 50 = h\sqrt{3}$

$\Rightarrow h+50 = h\sqrt{3}$

$\Rightarrow h+50 = h\sqrt{3}$ [from eq. (i)]

$\Rightarrow h = \frac{50}{\sqrt{3}-1}$ m

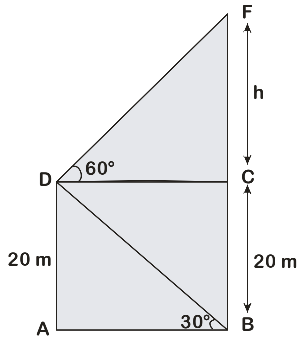
Sol. 10. (d)

In ΔABC, $\tan 30° = \frac{AD}{AB}$

$\Rightarrow \frac{1}{\sqrt{3}} \frac{AD}{AB}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{20}{AB}$

$\Rightarrow AB = 20\sqrt{3}m$



In ΔDCF, $\tan 60° = \frac{h}{DC} \Rightarrow \sqrt{3} = \frac{b}{DC}$

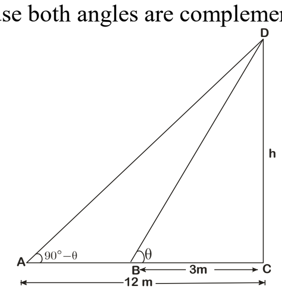
$\Rightarrow h = (\sqrt{3})(20\sqrt{3})$ ($\because$ AB = DC = $20\sqrt{3}$)

$\Rightarrow$h = 60m

$\therefore$ Height of tower, BF = 60 + 20 = 80m

Sol. 11. (b)

Let the height of the tower be *h* m and $\angle$CBD = θ then $\angle$ $\angle$DAC = 90° = θ
(Because both angles are complementary)



$\therefore$ In ΔABCD,

$\tan\theta = \frac{CD}{BC} \Rightarrow \tan\theta = \frac{h}{3}$

Now, in ΔACD

$\tan(90°-\theta) = \frac{CD}{AC} \Rightarrow \cot\theta = \frac{h}{12}$

$\tan\theta = \frac{h}{12}$

h tan θ = 12, put the value of tan θ

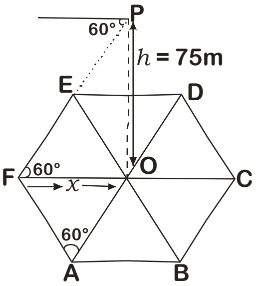
$h \times \frac{h}{13} = 12$

$h^2 = 36 \quad \therefore h = 6$

Then, height of tower = 6m.

Sol. 12. (c)

Let OP be the height of the tower



x = Distance between O and F.
In ΔFOP,

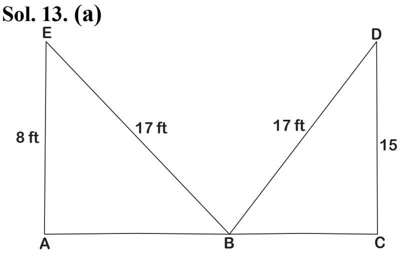
$$\tan 60° = \frac{75}{x} \Rightarrow x = \frac{75}{\sqrt{3}} = 25\sqrt{3}\text{m}$$

In regular hexagon Δ OEF, OED ... etc., are equilateral triangles.

$\therefore$ OF = AF = AB = BC = CD = DE = EF = $25\sqrt{3}$ m = side of hexagon

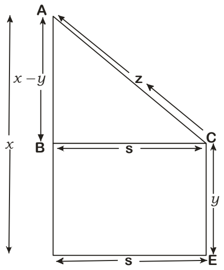
$\therefore$ Length of hexagon = $25\sqrt{3}$ m

Sol. 13. (a)



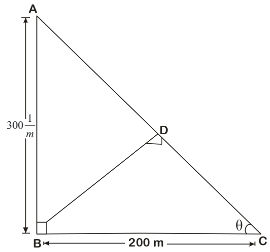
In Δ ABE,
$BE^2 = AE^2 + AB^2 \Rightarrow AB^2 = 17^2 - 8^2$
AB2= 289 – 64 = 225 $\Rightarrow$ AB = $\sqrt{225}$
AB = 15ft
In ΔBCD
BD2 = $BC^2 + CD^2 \Rightarrow BC^2 = 17^2 - 15^2$
= 289 – 225 = 64
$\Rightarrow$ BC = 8ft
$\therefore$ Width of the street = AB + BC = 15+8 = 23ft

Sol. 14. (d)



Here, BC is the shortest distance
In Δ ABC, $AB^2 + BC^2 = AC^2$ (Use Pythagoras theorem)
$\Rightarrow z^2 = (x-y)^2 + s^2$
$\Rightarrow z^2 = x^2 + y^2 - 2xy + s^2$
$\Rightarrow s^2 = z^2 - x^2 - y^2 + 2xy$

Sol. 15. (c)



Let a person be at point C observe a tower in west direction at B. After walking some distance along road he observed same tower in south direction. Let angle C be = θ

In Δ ABC

$\tan\theta \ \frac{300}{200} = \frac{3}{2} = 1$

h= $\sqrt{(3)^2 + (2)^2}$

$= \sqrt{9+4} = \sqrt{13}$

sin θ= $\frac{3}{\sqrt{13}}$

InΔBDC

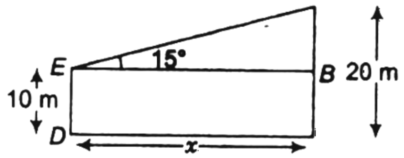
sinθ= $\frac{P}{h} = \frac{BD}{200}$

$\therefore$BD=200 sinθ

=200× $\frac{3}{\sqrt{13}} = \frac{600}{\sqrt{13}}$ m

Sol. 16. (b)

Let CD be the distance between two poles= xm
Here, AC=20 m and ED= 10 m



$\therefore$ AB=AC- BC=AC-ED=10 m
Now, in ΔAEB,

$$\tan 15^0 = \frac{AB}{BE}$$

$$\Rightarrow \tan\left(45^0 - 30^0\right) = \frac{10}{BE}$$

$$\left[\because \qquad \cdot B) = \frac{\tan A - \tan B}{1 + \tan A \cdot \tan B}\right]$$

$$\Rightarrow \frac{\tan 45^0 - \tan 30^0}{1 + \tan 45^0 \tan 30^0} = \frac{10}{BE} = \frac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}}$$

$$\Rightarrow \quad \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{10}{BE}$$

$$\Rightarrow \quad BE = 10\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right) = \frac{10 \times \left(\sqrt{3}+1\right)^2}{2}$$

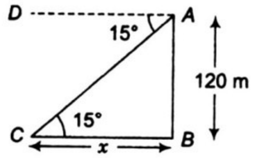
$$= 5\left(3 + 1 + 2\sqrt{3}\right) = 5\left(4 + 2 \times 1.73\right) \left[\because \qquad 732\right]$$

$$= 5\left(4 + 3.46\right)$$

$$\therefore \quad x = CD = BE = 5 \times 7.46 = 37.3m$$

Sol. 17. (c)

Let the distance between the boat from the lighthouse= xm



In ΔACB

$$\tan \ 15^0 = \frac{AB}{BC}$$

$$\Rightarrow \tan\left(45^0 - 30^0\right) = \frac{120}{BC}$$

$$\Rightarrow \frac{\tan 45^0 - \tan 30^0}{1 + \tan 45^0 \tan 30^0} = \frac{120}{BC} = \frac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}}$$

$$\Rightarrow \quad \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{120}{BC}$$

$$\therefore \ x = BC = 120\left(\frac{\left(\sqrt{3}+1\right)^2}{3-1}\right)$$

$$= 60\left(3 + 1 + 2\sqrt{3}\right)$$

$$= 60\left(4 + 2 \times 1.73\right) \qquad \left[\because \qquad 732\right]$$

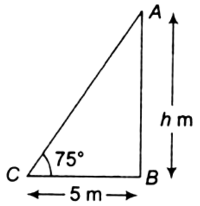
$= 60 \times 7.46 = 447.6$ m

= 448 m (approx)

$\therefore$ Because 448 m (approx) is near about 444m. So, option (c) is correct.

Sol. 18. (c)

Let AB be the height of the flag post =h m



In ΔACB,

$$\tan 75^0 = \frac{AB}{BC} = \frac{h}{5}$$

$$\Rightarrow \tan\left(45^0 + 30^0\right) = \frac{\tan 45^0 + \tan 30^0}{1 - \tan 45^0 \tan 30^0} = \frac{h}{5}$$

$$\left[\because \qquad \cdot B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}\right]$$

$$\Rightarrow \frac{1+\sqrt{3}}{\sqrt{3}-1} = \frac{h}{5}$$

$$\therefore h = \frac{\left(\sqrt{3}+1\right)^2}{\left(\sqrt{3}\right)^2 - (1)^2} \times 5 = 5\left(\frac{3 + 1 + 2\sqrt{3}}{3-1}\right)$$

$$= 5\left(2 + \sqrt{3}\right) \qquad \left[\because \qquad 732\right]$$

=5×3.732 = 18.660 = 19 m(approx)

OFFICIUM SUPREMUM

NDA PYQ

1. A man standing on the bank of a river observes that the angle of elevation of the top of a tree just on the opposite bank is 60^0. The angle of elevation is 30^0 from a point at a distance y m from the bank. What is the height of the tree?

(a) y m (b) 2y m

(c) $\frac{\sqrt{3}y}{2}m$ (d) $\frac{y}{2}m$

[NDA (I) - 2011]

2. A tower of height 15 m stands vertically on the ground. From a point on the ground the angle of elevation of the top of the tower is found to be 30^0. What is the distance of the point from the foot of the tower?

(a) $15\sqrt{3}$ m (b) $10\sqrt{3}$ m

(c) $5\sqrt{3}$ m (d) 30 m

[NDA (II) - 2011]

3. At a point 15 m away from the base of a 15 m high house, the angle of elevation of the top is

(a) 90^0 (b) 60^0

(c) 45^0 (d) 30^0

[NDA (II) - 2011]

4. A vertical tower stands on a horizontal plane and is surmounted by a vertical flag staff of elevation of the bottom of the flag staff is β and that of the top is α. What is the height of the tower?

(a) $\frac{h\tan\beta}{\tan\alpha-10\beta}$ (b) $\frac{h\tan\beta}{\tan\alpha+\tan\beta}$

(c) $\frac{h\cos\beta}{\tan\alpha-\tan\beta}$ (d) $\frac{h}{\cos(\alpha-\beta)}$

[NDA (II) - 2011]

5. An aeroplane flying at a height of 300 m above the ground passes vertically above another plane at an instant when the angles of elevation of two planes from the same point on the ground are 60^0 and 45^0, respectively. What is the height of the lower plane from the ground?

(a) 50 m (b) $\frac{100}{\sqrt{3}}$ m

(c) $100\sqrt{3}$ m (d) $150\left(\sqrt{3}+1\right)$ m

[NDA (II) - 2011]

6. What is the angle subtended by 1 m pole at distance 1 km on the ground in sexagesimal measure?

(a) $\frac{9}{50\pi}$ degree (b) $\frac{9}{5\pi}$ degree

(d) 3.4 min (d) 3.5 min

[NDA (I) – 2012]

7. Two poles are 10 m and 20 m high. The line joining their tips makes an angle of 15^0 with the horizontal. What is the distance between the poles?

(a) $10\left(\sqrt{3}-1\right)$ m (b) $5\left(4+2\sqrt{3}\right)$ m

(c) $20\left(\sqrt{3}+1\right)$ (d) $10\left(\sqrt{3}+1\right)$ m

[NDA (I) - 2012]

8. The angle of elevation of a tower at a level ground is 30^0. The angle of elevation becomes θ when moved 10m towards the tower. If the height of tower is $5\sqrt{3}$ m, then what is the value of θ ?

(a) 45^0 (b) 60^0

(c) 75^0 (d) None

[NDA (I) - 2012]

9. From the top of a building of height h m, the angle of depression of an object on the ground is θ. What is the distance (in m) of the object from the foot of the building?

(a) h cot θ (b) h tan θ

(c) h cos θ (d) h sin θ

[NDA (I) - 2012]

10. The angle of elevation of the tip of a flag staff from a point 10 m due South of its base is 60°. What is the height of the flag staff correct to nearest meter?

(a) 15 m (b) 16 m

(c) 17 m (d) 18 m

[NDA (I) - 2012]

11. The top of a hill observed from the top and bottom of a building of height h is at angles of elevation α and β, respectively. The height of the hill is

(a) $\frac{h\cot\beta}{\cot\beta-\cot\alpha}$ (b) $\frac{h\cot\alpha}{\cot\alpha-\cot\beta}$

(c) $\frac{h\cot\alpha}{\tan\alpha-\tan\beta}$ (d) None of these

[NDA (II) - 2012]

12. From the top of a lighthouse 70 high with its base at sea level, the angle of depression of a boat is 15^0. The distance of the boat from the foot of the lighthouse is

(a) $70\left(2-\sqrt{3}\right)m$ (b) $70\left(2+\sqrt{3}\right)m$

(c) $70\left(3-\sqrt{3}\right)m$ (d) $70\left(3+\sqrt{3}\right)m$

[NDA (II) - 2012]

13. The angle of elevation of the top of a tower of height H from the foot of another tower in the same plane is 60^0 and the angle of elevation of the top of the second tower from the foot of the first tower is 30^0. If h is the height of the other tower, then which one of the following is correct?

(a) H=2h (b) $H=\sqrt{3}\ h$

(c) H = 3h (d) None of these

[NDA (I) - 2013]

14. A man walks 10 m towards a lamp post and notices that the angle of elevation of the top of the post increases from 30^0 to 45^0 the height of the lamp post is

(a) 10 m (b) $\left(5\sqrt{3}+5\right)m$

(c) $\left(5\sqrt{3}-5\right)m$ (d) $\left(10\sqrt{3}+10\right)m$

[NDA (I) - 2013]

15. The shadow of a tower standing on a level plane is found to be 50 m longer when the Sun's elevation is 30^0 than when it is 60^0. The height of the tower is

(a) 25 m (b) $25\sqrt{3}\ m$

(c) 50 m (d) none of these

[NDA (I) - 2013]

16. If the angles of elevation of the top of a tower from two places situated at distances 21 m and x m from the base of the tower are 45^0 and 60^0 respectively, then what is the value of x?

(a) $7\sqrt{3}\ m$ (b) $7-\sqrt{3}\ m$

(c) $7+\sqrt{3}\,m$ (d) 14

[NDA (II) - 2013]

17. A person standing on the bank of a river observes that the angle subtended by a tree on the opposite of bank is 60^0 when he returns 40 m from the bank, he finds the angle to be 30^0. What is the breadth of the river?

(a) 60 m (b) 40 m
(c) 30 m (d) 20 m

[NDA (II) - 2013]

18. From an aeroplane above a straight road the angles of depression of two positions at a distance 20 m apart on the road are observed to be 30^0 and 45^0. The height of the aeroplane above the ground is

(a) $10\sqrt{3}\,m$ (b) $10\left(\sqrt{3}-1\right)m$
(c) $10\left(\sqrt{3}+1\right)m$ (d) $20\ m$

[NDA (I) - 2014]

19. A lamp post stands on a horizontal plane. From a point situated at a distance 150 m from its foot, the angle of elevation of the top is 30^0. What is the height of the lamp post?

(a) 50 m (b) $50\sqrt{3}\ m$
(c) $\frac{50}{\sqrt{3}}\ m$ (d) $100\ m$

[NDA (II) - 2014]

20. The angle of elevation of the top of a tower from a point 20 m away from its base is 45^0. What is the height of the tower?

(a) 10 m (b) 20 m
(c) 30 m (d) 40 m

[NDA (I) - 2015]

21. The angles of elevation of the top of a tower standing on a horizontal plane from two points on a line passing through the foot of the tower at distances 49 m and 36 m are 43^0 and 47^0 respectively. What is the height of the tower?

(a) 40 m (b) 42 m
(c) 45 m (d) 47 m

[NDA (I) - 2015]

22. Two poles are 10 m and 20 m high. The line joining their tops makes an angle of 15^0 with the horizontal. The distance between the poles is approximately equal to

(a) 36.3 m (b) 37.3 m
(c) 38.3 m (d) 39.3 m

[NDA (II) - 2015]

23. A vertical tower standing on a leveled field is mounted with a vertical flag staff of length 3 m. From a point on the field, the angles of elevation of the bottom and tip of the flag staff are 30^0 and 45^0, respectively. Which one of the following gives the best approximation to the height of the tower?

(a) 3.90 m (b) 4.00 m
(c) 4.10 m (d) 4.25 m

[NDA (II) - 2015]

24. The top of a hill when observed from the top and bottom of a building of height h is at angles of elevation p and q respectively. What is the height of the hill?

(a) $\frac{h\cot q}{\cot q-\cot p}$ (b) $\frac{h\cot p}{\cot p-\cot q}$
(c) $\frac{2h\tan p}{\tan p-\tan q}$ (d) $\frac{2h\tan q}{\tan q-\tan p}$

[NDA (II) - 2016]

25. A moving boat is observed from the top of a cliff of 150 m height. The angle of depression of the boat changes from 60^0 to 45^0 in 2 minutes. What is the speed to the boat in metres per hour?

(a) $\frac{4500}{\sqrt{3}}$ (b) $\frac{4500\left(\sqrt{3}-1\right)}{\sqrt{3}}$
(c) $4500\sqrt{3}$ (d) $\frac{4500\left(\sqrt{3}+1\right)}{\sqrt{3}}$

[NDA (II) - 2016]

26. From the top of a lighthouse, 100 m high, the angle of depression of a boat is $\tan^{-1}\left(\frac{5}{12}\right)$. What is the distance between the boat and the lighthouse?

(a) 120 m (b) 180 m
(c) 240 m (d) 360 m

[NDA (I) - 2017]

27. The angle of elevation of a stationary cloud from a point 25 m above a lake is 15^0 and the angle of depression of its image in the lake is 45^0. The height the cloud above the lake level is

(a) 25 m (b) $25\sqrt{3}\,m$
(c) 50 m (d) $50\sqrt{3}\ m$

[NDA (II) - 2017]

28. The angles of elevation of the top of a tower form the top and foot of a pole are respectively 30^0 and 45^0 is h_T is the height of the tower and h_P is the height of the pole, then which of the following are correct?

1. $\frac{2h_p h_T}{3+\sqrt{3}}=h_P^2$
2. $\frac{h_T-h_p}{\sqrt{3}+1}=\frac{h_P}{2}$
3. $\frac{2\left(h_p+h_T\right)}{h_p}=4+\sqrt{3}$

Select the correct answer using the code given below:

(a) 1 and 3 only (b) 2 and 3 only
(c) 1 and 2 only (d) 1, 2 and 3

[NDA (II) - 2017]

29. If a flag-staff of 6 m height placed on the top of a tower throws a shadow of $2\sqrt{3}$ m along the ground, then what is the angle that the sun makes with the ground?

(a) 60^0 (b) 45^0
(c) 30^0 (d) 15^0

[NDA (I) - 2018]

30. A spherical balloon of radius r subtends an angle α at the eye of an observer, while the angle of elevation of its centre is β. What is the height of the centre of the balloon (neglecting the height of the observer)?

(a) $\frac{r\sin\beta}{\sin\left(\frac{\alpha}{2}\right)}$ (b) $\frac{r\sin\beta}{\sin\left(\frac{\alpha}{4}\right)}$
(c) $\frac{r\sin\left(\frac{\beta}{2}\right)}{\sin\alpha}$ (d) $\frac{r\sin\alpha}{\sin\left(\frac{\beta}{2}\right)}$

[NDA (I) - 2018]

31. The top of a hill observed from the top and bottom of a building height h is at angles of elevation $\frac{\pi}{6}$ and $\frac{\pi}{3}$ respectively. What is the height of the hill?

(a) $2h$ (b) 3h/2

(c) h (d) h/2

[NDA (II) - 2018]

32. Balloon is directly above one end of bridge. The angle of depression of the other end of the bridge from the balloon is 48^0. If the height of the balloon above the bridge is 122 m, then what is the length of the bridge?

(a)122 sin 48^0 m (b) 122 tan 42^0 m

(c) 122 cos 48^0 m (d) 122 tan 48^0 m

[NDA (II) - 2018]

33. The angle of elevation of a tower of height h from a point A due south of it is x and from a point B due east of A is y. If AB = z, then which one of the following is correct?

(a) $h^2 (\cot^2 y - \cot^2 x) = z^2$ (b) $z^2 (\cot^2 y - \cot^2 x) = h^2$

(c) $h^2 (\tan^2 y - \tan^2 x) = z^2$ (d) $h^2 (\tan^2 y - \tan^2 x) = h^2$

[NDA (I) - 2019]

34. A ladder 9m long reaches a point 9 m below the top of a vertical flagstaff. From the foot of the ladder, the elevation of the top of the flagstaff is 60°. What is the height of the flag staff?

(a) 9 m (b) 10.5 m

(c) 13.5 m (d) 15 m

[NDA (II) - 2019]

35. A ladder 6m long reaches a point 6 m below the top of a vertical flagstaff. From the foot of the ladder, the elevation of the top of the flagstaff is 75°. What is the height of the flag staff?

(a) 12 m (b) 9 m

(c) $(6 + \sqrt{3})$ m (d) $(6 + 3\sqrt{3})$ m

[NDA (I) - 2021]

36. The shadow of a tower is found to be x meter longer, when the angle of elevation of the sun changes from 60° to 45°. If the height of the tower is $5(3 + \sqrt{3})$m, then what is x equal to ?

(a) 8 m (b) 10 m

(c) 12 m (d) 15 m

[NDA (I) - 2021]

37. A vertical tower stands on a horizontal plane and is surmounted by a vertical flagstaff of height h. At a point on the plane the angles of elevation of the bottom and top of the flagstaff are θ and 2θ respectively. What is the height of the tower?

(a) hcosθ (b) hsinθ

(c) hcos2θ (d) hsin2θ

[NDA (I) - 2022]

38. The shadow of a tower becomes x metre longer, when the angle of elevation of sun changes from 60° to θ. If the height of the tower is $\sqrt{3}x$ meter, then which one of the following is correct?

(a) $0 < \theta < 30°$ (b) $30° < \theta < 45°$

(c) $45° < \theta < 60°$ (d) $60° < \theta < 90°$

[NDA (I) - 2022]

Consider the following for the next three (03) items that follows:

There are two points P and Q due south of a leaning tower, which leans towards north. P is at a distance x and Q is at a distance y from the foot of the tower (x > y). The angles of elevation of the top of the tower from P and Q are 15° and 75° respectively.

39. At what height is the top of the tower above the ground level?

(a) $\frac{x-y}{2\sqrt{3}}$ (b) $\frac{x-y}{4\sqrt{3}}$

(c) $\frac{x-y}{4}$ (d) $\frac{x-y}{2}$

[NDA 2022 (II)]

40. If θ is the inclination of the tower to the horizontal, then what is cotθ equal to?

(a) $2+\frac{\sqrt{3}(x-y)}{x+y}$ (b) $2-\frac{\sqrt{3}(x-y)}{x+y}$

(c) $2+\frac{\sqrt{3}(x+y)}{x-y}$ (d) $2-\frac{\sqrt{3}(x+y)}{x-y}$

[NDA 2022 (II)]

41. What is the length of the tower?

(a) $\frac{x-y}{2\sqrt{3}}\sqrt{1+\left\{2+\frac{\sqrt{3}(x+y)}{x-y}\right\}^2}$

(b) $\frac{x-y}{2\sqrt{3}}\sqrt{1+\left\{2-\frac{\sqrt{3}(x+y)}{x-y}\right\}^2}$

(c) $\frac{x-y}{4\sqrt{3}}\sqrt{1+\left\{2+\frac{\sqrt{3}(x+y)}{x-y}\right\}^2}$

(d) $\frac{x-y}{4\sqrt{3}}\sqrt{1+\left\{2-\frac{\sqrt{3}(x+y)}{x-y}\right\}^2}$

[NDA 2022 (II)]

Consider the following for the next two (02) items that follow:

A flagstaff 20m long standing on a pillar 10m high subtends an angle $\tan^{-1}(0.5)$ at a point P on the ground. Let θ be the angle subtended by the pillar at this point P.

42. If x is the distance of P from bottom of the pillar, then consider the following statements:

1.x can take two values which are in the ratio 1:3

2.x can be equal to height of the flagstaff

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) both 1 and 2 (d) neither 1 nor 2

[NDA – 2023 (1)]

43. What is a possible value of tanθ?

(a) 3/4 (b) 2/3

(c) 1/3 (d) 1/4

[NDA – 2023 (1)]

Consider the following for the two (02) items that follow:

The top (M) of a tower is observed from three points P, Q and R lying in a horizontal straight line which passes directly along the foot (N) of the tower. The angles of elevations of M from P, Q and R are 30°, 45°, and 60° respectively. Let PQ = a and QR = b.

44. What is PN equal to?

(a) $\left(\frac{3-\sqrt{3}}{2}\right)a$ (b) $\left(\frac{3+\sqrt{3}}{2}\right)a$

(c) $\left(\frac{3-\sqrt{3}}{4}\right)a$ (d) $\left(\frac{3+\sqrt{3}}{4}\right)a$

[NDA – 2025 (1)]

45. What is MN equal to?

(a) $\left(\frac{3+\sqrt{3}}{2}\right)b$ (b) $\left(\frac{3-\sqrt{3}}{2}\right)b$

(c) $\left(\frac{3-\sqrt{3}}{4}\right)b$ (d) $\left(\frac{3+\sqrt{3}}{4}\right)b$

[NDA – 2025 (1)]

46. A man at M, standing 100 m away from the base (P) of a chimney of height 50 m, observes the angle of elevation of the highest point (Q) of the smoke to be 45°. The highest point of the chimney is at R. Further P, R and Q are in a straight line and the straight line is perpendicular to PM. What is the angle ∠RMQ equal to?

(a) $\tan^{-1}\left(\frac{1}{2}\right)$ (b) $\tan^{-1}\left(\frac{1}{3}\right)$

(c) $\tan^{-1}\left(\frac{2}{3}\right)$ (d) $\tan^{-1}\left(\frac{3}{4}\right)$

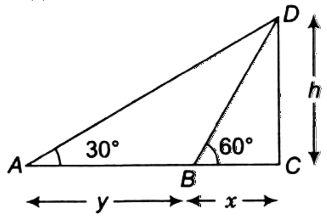
[NDA – 2025 (1)]

ANSWER KEY

1.	c	**2.**	a	**3.**	c	**4.**	a	**5.**	c	**6.**	a	**7.**	b	**8.**	b	**9.**	a	**10.**	c
11.	b	**12.**	b	**13.**	c	**14.**	b	**15.**	b	**16.**	a	**17.**	d	**18.**	c	**19.**	b	**20.**	b
21.	b	**22**	b	**23.**	c	**24.**	b	**25.**	b	**26.**	c	**27.**	b	**28.**	c	**29.**	a	**30.**	a
31.	b	**32.**	b	**33.**	a	**34.**	c	**35.**	d	**36.**	b	**37.**	c	**38.**	b	**39.**	a	**40.**	d
41.	b	**42.**	a	**43.**	c	**44.**	b	**45.**	a	**46.**	b								

Solutions

Sol.1. (c)



In ΔACD, $\tan 30^0 = \frac{CD}{AC} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{x+y}$

$\Rightarrow \quad x + y = h\sqrt{3}$ (i)

And in ΔCBD, $\tan 60^0 = \frac{CD}{BC}$

$\Rightarrow \sqrt{3} = \frac{h}{x} \Rightarrow x = \frac{h}{\sqrt{3}}$ …(iii)

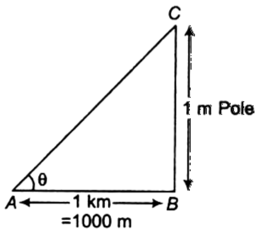
From Eqs. (i) and (ii),

$\frac{h}{\sqrt{3}} + y = h\sqrt{3} \Rightarrow y = h\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right) = \frac{2h}{\sqrt{3}}$

$\therefore \quad h = \frac{\sqrt{3}y}{2} m$

Sol.2. (a)

Let $\angle CAB = \theta$



Then, $\tan\theta = \frac{BC}{AB} = \frac{1}{1000}$ radian

$\Rightarrow \quad \tan\theta = 0.001 radian$

$\therefore \quad \theta = \tan^{-1}(0.001) radian$

=9.9999 X 10^{-4} radian

$9.999 \times 10^{-4} \times \frac{180}{\pi} \deg ree$

= 0.057324 degree

$= \frac{9}{50\pi}$ degree

Sol.3. (a)

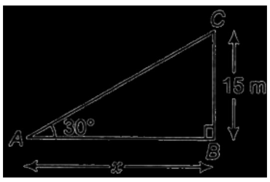
Let AB be the distance of point from the foot of the tower = xm

Then, in ΔBAC,

$\tan 30^0 = \frac{BC}{AB} = \frac{15}{x}$

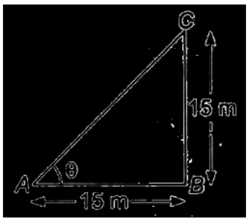
$\Rightarrow \frac{1}{\sqrt{3}} = \frac{15}{x}$

$\therefore x = 15\sqrt{3}m$



Sol.4. (c)

Let the angle of elevation = θ

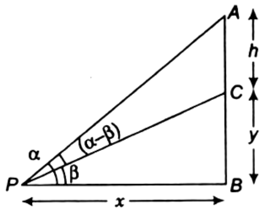


In ΔBAC ,

$\tan\theta = \frac{BC}{AB} = \frac{15}{15} = 1 \Rightarrow \theta = 45^0$

Sol.5. (a)

let the height of the tower = y = BC



Now, in ΔCPB

$\tan\beta = \frac{y}{x}$ …(i)

And in ΔAPB, $\tan a = \frac{AB}{BP} = \frac{AC + BC}{BP}$

$\Rightarrow \quad \tan\alpha = \frac{h+y}{x}$

$\Rightarrow \quad y + h = x\tan\alpha$

$\Rightarrow \quad$ y+h= $\frac{y}{\tan\beta}$ ×tanα [from Eq. (i)]

$\Rightarrow \quad y\left(1 - \frac{\tan\alpha}{\tan\beta}\right) = -h$

$\Rightarrow \quad y(\tan\alpha - \tan\beta) = h\tan\beta$

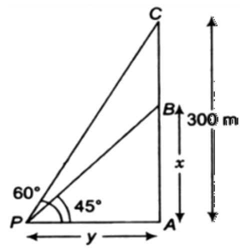
$\therefore$ Height of the tower,

$y = \frac{h\tan\beta}{(\tan\alpha - \tan\beta)}$

Sol.6. (c)

Let AB be the height of the lower plane from the ground = x m

And PA = y m



Now, in ΔAPB

$\tan 45^0 = \frac{x}{y} = \frac{AB}{AP} = 1 \Rightarrow x = y$ …(i)

and in ΔAPC

$\tan 60^0 = \frac{AC}{AP} = \frac{300}{y} = \sqrt{3}$

$\Rightarrow \quad y\frac{300}{\sqrt{3}}$

$\therefore x = \frac{300}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{300\sqrt{3}}{3}$ [from Eq.(i)]

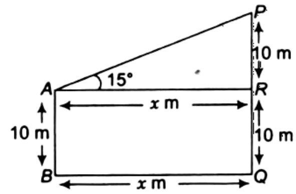
$= 100\sqrt{3}m$

Sol.7. (b)

Let x m be the distance between the two poles.

In ΔPAR, $\tan 15^0 = \frac{10}{x} = \tan(45^0 - 30^0)$

$\Rightarrow \frac{10}{x} = \frac{\tan 45^0 - 30^0}{1 + \tan 45^0 \tan - 30^0} = \frac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}} = \frac{\sqrt{3}-1}{\sqrt{3}+1}$



$\Rightarrow \frac{10}{x} = \frac{\sqrt{3}-1}{\sqrt{3}+1} \cdot \frac{\sqrt{3}+1}{\sqrt{3}-1} = 2 - \sqrt{3}$

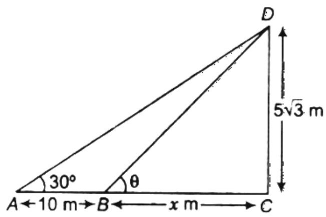
$\therefore x = \frac{10}{2-\sqrt{3}} \cdot \frac{2+\sqrt{3}}{2+\sqrt{3}} = 10(2+\sqrt{3}) = 5(4+2\sqrt{3})m$

So, the required distance between the pole is $5(4+2\sqrt{3})$ m.

Sol.8. (b)

After moving towards the tower, let the remaining distance between the point where angle made θ , is x m.

InΔ DBC, $\tan\theta = \frac{5\sqrt{3}}{x}$ …(i)



Again, in ΔDAC ,

$\tan 30^0 = \frac{5\sqrt{3}}{10+x} = \frac{1}{\sqrt{3}}$

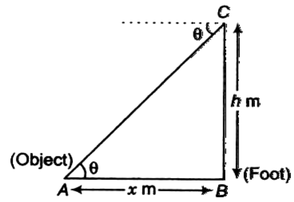
$\Rightarrow 10 + x = 15$

From Eq. (i)

$\tan 30^0 = \frac{5\sqrt{3}}{5} = \sqrt{3} = \tan 60^0 \Rightarrow \theta = 60^0$

Sol.9. (a)

Let AB be the distance between object and foot of the building = x m



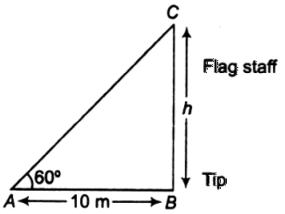
In ΔABC, $\tan\theta = \frac{h}{x}$

$\therefore\ x = h\cot\theta$

So, the required distance is hcotθ

Sol.10. (c)

Let CB be flag staff of height hm.



In ΔBAC

$\tan 60^0 = \frac{h}{10}$

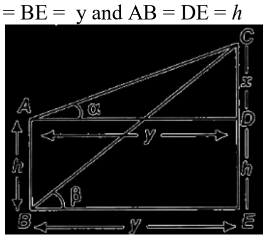
$\therefore h = 10\tan 60^0 = 10.\sqrt{3}\quad \left[\because \quad = \sqrt{3}\right]$

$= 10(1.732)$

=17.32 m =17 m (approx)

Sol.11. (b)

Let AD = BE = y and AB = DE = h



Now, in ΔCAD,

$\tan\alpha = \frac{x}{y} \Rightarrow\ y = x\cot\alpha$ ..(i)

Again, in ΔCBE, $\tan\beta = \frac{x+h}{y}$

$\Rightarrow\quad y = (x+h)\cot\beta$ ….(ii)

From Eqs. (i) and (ii)

$x\cot\alpha = (x+h)\cot\beta$

$\Rightarrow x(\cot\alpha - \cot\beta) = h\cot\beta$

$\Rightarrow x\cot\alpha = x\cot\beta + h\cot\beta$

$\Rightarrow x = \frac{h\cot\beta}{\cot\alpha - \cot\beta}$ …(iii)

$\because$ Required height

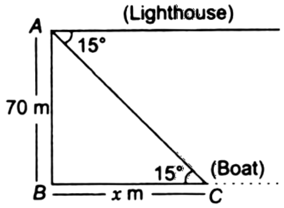
$(CE) = CD + DE = x + h$

$= \frac{h\cot\beta}{\cot\alpha - \cot\beta} + h$

$= \frac{h\cot\alpha}{\cot\alpha - \cot\beta}$

Sol.12. (b)

Let the distance of the boat from the foot of lighthouse is h m.



$\Rightarrow\ \tan(45^0 - 30^0) = \frac{70}{x}$

$\Rightarrow \frac{1-\frac{1}{\sqrt{3}}}{1+\frac{1}{\sqrt{3}}} = \frac{70}{x} \Rightarrow \frac{\sqrt{3}-1}{1+\sqrt{3}} = \frac{70}{x}$

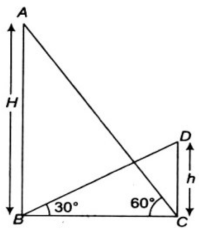
$\therefore x = 70\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)\cdot\left(\frac{\sqrt{3}+1}{\sqrt{3}+1}\right)$

…[by rationalization]

$= \frac{70\left(3+1+2\sqrt{3}\right)}{3-1} = 70\left(2+\sqrt{3}\right)$

Hence, the required distance $70\left(2+\sqrt{3}\right)m$

Sol.13. (c)



In Δ ABC,

$\tan 60^0 = \frac{H}{BC} \Rightarrow \sqrt{3} = \frac{H}{BC} \Rightarrow BC = \frac{H}{\sqrt{3}}$ .....(i)

In ΔDBC,

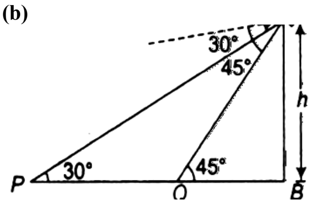
$\tan 30^0 = \frac{H}{BC} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{BC}$

$\Rightarrow BC = h\sqrt{3}$ ....(ii)

From Eqs. (i) and (ii), $\frac{H}{\sqrt{3}} = h\sqrt{3}$

H = 3h

Sol.14. (b)



Given that and $\angle AQB = 45^0$

Now, in ΔAQB,

$\tan 45^0 = \frac{AB}{QB} = \frac{h}{x} \Rightarrow 1 = \frac{h}{x}$

$\Rightarrow\quad x = h$ … (i)

and in ΔAPB,

$\tan 30^0 = \frac{AB}{PB} = \frac{h}{PQ+QB}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{10+x} \Rightarrow \sqrt{3}\,h = 10 + x = 10 + h$

[from Eq. (i)]

$\Rightarrow\left(\sqrt{3}\ h - h\right) = 10 \Rightarrow h\left(\sqrt{3}-1\right) = 10$

$\therefore\ h = \frac{10}{\sqrt{3}-1}\cdot\frac{\sqrt{3}+1}{\sqrt{3}+1} = 10$

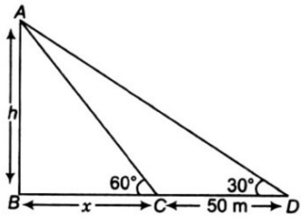
….[by rationalization]

$h = \frac{10\left(\sqrt{3}+1\right)}{3-1} = \frac{10\left(\sqrt{3}+1\right)}{2} = 5\sqrt{3}+5$

hence, the required height is $5\left(\sqrt{3}+1\right)$ m

Sol.15. (b)

Let shadow of a tower made by a Sun at an $\angle 60°$ is x m. Then, by given condition, the shadow of a tower made by a Sun at an $\angle 30°$ is 50 m longer than at an angle 60^0 i.e. (50 + x)m.



$\Rightarrow x = \frac{h}{\sqrt{3}}m$

and in ΔADB,

$\tan 30^0 = \frac{h}{x+50} = \frac{1}{\sqrt{3}} \Rightarrow \sqrt{3}h = x + 10$

$\Rightarrow \sqrt{3}h = \frac{h}{\sqrt{3}} + 50$

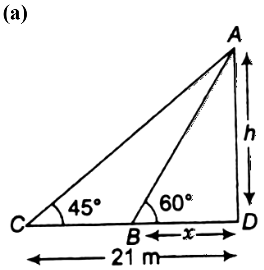
…[from Eq. (i)]

$\Rightarrow\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right)h = 50 \Rightarrow \frac{(3-1)}{\sqrt{3}} = 50$

$\Rightarrow 2h = 50\sqrt{3}$

$\therefore h = 25\sqrt{3}m$

Sol.16. (a)



In ΔACD, $\tan 45^0 = \frac{h}{21} \Rightarrow 1 = \frac{h}{21}$ …(i)

$\therefore\quad h = 21m$

Again in ΔABD,

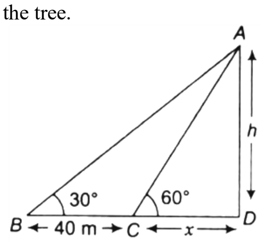
$\tan 60^0 = \frac{h}{x} \Rightarrow \sqrt{3} = \frac{h}{x}$

$\therefore\quad x = \frac{21}{\sqrt{3}} = 7\sqrt{3}m$

[from Eq.(i)]

Sol.17. (d)

Let x m be the breadth of the river and h m be the height of the tree.



Now, in ΔACD,

$\tan 60^0 = \frac{h}{x} \Rightarrow \sqrt{3} = \frac{h}{x}$

$\therefore h = \sqrt{3}x$

Again, in ΔABD,

$\tan 30^0\ \frac{h}{x+40} = \frac{1}{\sqrt{3}}$

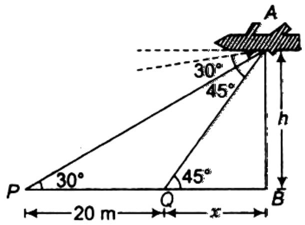
$\Rightarrow h\sqrt{3} = x + 40$

$\Rightarrow \sqrt{3}\,x\cdot\sqrt{3} = x + 40$

$\Rightarrow 3x - x = 40$

So, the breadth of the rivers is 20 m.

Sol.18. (c)
Let the height of the aeroplane above the ground is hm and QB= xm.



Given that and $\angle AQB = 45^0$

Now, in ΔAQB,

$\tan 45^0 = \frac{AB}{QB} = \frac{h}{x} \Rightarrow 1 = \frac{h}{x}$

$\Rightarrow \quad x = h$ … (i)

and in ΔAPB,

$\tan 30^0 = \frac{AB}{PB} = \frac{h}{PQ+QB}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{20+x} \Rightarrow \sqrt{3}\,h = 20 + x = 20 + h$

[from Eq. (i)]

$\Rightarrow \left(\sqrt{3}\ h - h\right) = 20 \Rightarrow h\left(\sqrt{3}-1\right) = 20$

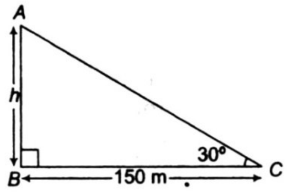
$\therefore h = \frac{20}{\sqrt{3}-1} \cdot \frac{\sqrt{3}+1}{\sqrt{3}+1} = 20$

...[by rationalization]

$= \frac{20\left(\sqrt{3}+1\right)}{\left(3-1\right)} \cdot \frac{20\left(\sqrt{3}+1\right)}{2} = 10\left(\sqrt{3}+1\right)$

hence, the required height is $10\left(\sqrt{3}+1\right)$ m.

Sol.19. (b)
Let AB be the lamp post of height h m and C is a point situated at a distance of 150 m from its foot B.



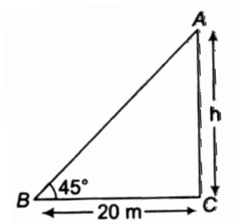
In ΔABC, we have

$\tan 30^0 = \frac{h}{150}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{150}$

$\therefore h = \frac{150}{\sqrt{3}} = \frac{150\times\sqrt{3}}{3} = 50\sqrt{3}m$

Sol.20. (b)
Let AC be tower of height h m.



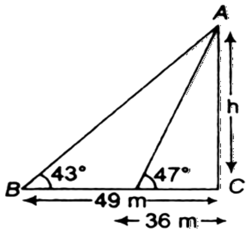
In ΔABC, $\tan 45^0 = \frac{h}{20} \Rightarrow h = 20m$

Sol.21. (b)
Let AC be the height of tower h m In

ΔACD, $\tan 47^0 = \frac{h}{36}$ ….(i)

And in ΔABD, $\tan 43^0 = \frac{h}{49}$ …(ii)

$\Rightarrow \quad \frac{1}{\cot 43^0} = \frac{h}{49} \Rightarrow \frac{49}{\cot 43^0} = h$



$\Rightarrow \quad h = \frac{49}{\cot\left(90-47^0\right)}$

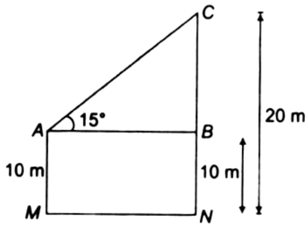
$\Rightarrow \quad h = \frac{49}{\cot 47^0}$ ….[from Eq.(i)]

$\Rightarrow \quad h = \frac{49}{\left(\frac{h}{36}\right)} \Rightarrow h^2 = 49\times 36$

$\therefore \quad h = 7\times 6 = 42\,m$

Sol.22. (b)
Here, CB = CN - BN = 20 - 10 = 10 m

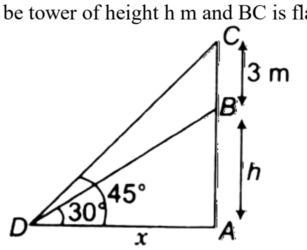
Now, in ΔABC, $\tan 15^0 = \frac{BC}{AB}$



$\Rightarrow \quad AB = \frac{BC}{\tan 15^0} = \frac{10}{2-\sqrt{3}} = 37.3m$

$\left[\because \quad = 2-\sqrt{3}\right]$

Sol.23. (c)
Let AB be tower of height h m and BC is flag.



In Δ DAB, tan 30^0=h/x

$\Rightarrow x = \sqrt{3}\ h$

Again, ΔDAC, tan $45^0 = \frac{h+3}{x}$

$\Rightarrow \quad x = h + 3$

$\Rightarrow h = x - 3 = \sqrt{3}\ h - 3$

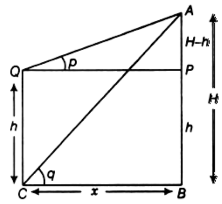
$\Rightarrow h\left(1-\sqrt{3}\right) = -3$

$\therefore h = \frac{3}{\sqrt{3}-1} = \frac{3\left(\sqrt{3}-1\right)}{2}$

$= \ 1.5\times 2.732 = 4.098m$

= 4.10 m (approx)

Sol.24. (b)
Let the height of hill be H and distance between building and hill be x.



Given, CQ=BP=h height of building

$\angle AQP = P$

and $\angle ACB = q$

In ΔABC, $\tan q \frac{H}{x}$

$\Rightarrow x \frac{H}{\tan q}$

In ΔAPQ,

$\tan p = \frac{H}{\tan q}$

$\Rightarrow x = \frac{H-n}{\tan p}$

From eqs. (i) and (ii) we get

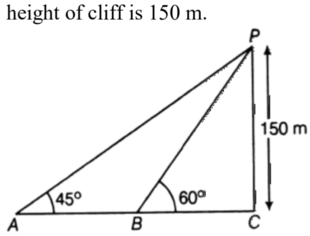
$\frac{H}{\tan q} = \frac{H-h}{\tan p}$

$\Rightarrow H \tan p = H \tan q - h \tan q$

$\Rightarrow H\left(\tan q - \tan p\right) = h \tan q$

$\therefore H = \frac{h \cot p}{\cot p - \cot q}$

Sol.25. (b)
Given, height of cliff is 150 m.



Let P is top of the cliff and its height PC=150m
Let A and B are the position of boat

In ΔPBC, $\tan 60^0 = \frac{PC}{BC}$

$\Rightarrow \quad \sqrt{3} = \frac{150}{BC}$

$\Rightarrow \quad BC = \frac{150}{\sqrt{3}}\times\frac{\sqrt{3}}{\sqrt{3}} = \frac{150\sqrt{3}}{3} = 50\sqrt{3}m$

In ΔPAC, $\tan 45^0 = \frac{PC}{AC} \Rightarrow 1 = \frac{150}{AC}$

$\Rightarrow AC = 150m$

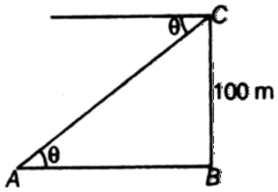
Now, distance traced in 2 min, AB=AC-BC=150-$50\sqrt{3}$

$\Rightarrow AB = 50\ \left(3-\sqrt{3}\right)$

∴ Speed of boat in metres per hour= $\frac{\text{Distance}}{\text{Time Taken hour}}$

$= \frac{50\left(3-\sqrt{3}\right)\times 60}{2} = \frac{4500\left(\sqrt{3}-1\right)}{\sqrt{3}}$

Sol.26. (c)
Let C be the top of a lighthouse



Then, $\tan\theta = \dfrac{100}{AB}$

$\Rightarrow \dfrac{5}{12} = \dfrac{100}{AB}$ [given, $\tan\theta = \dfrac{5}{12}$]

$\Rightarrow AB = \dfrac{12\times 100}{5}$

$\Rightarrow AB = 240\ m$

Sol.27. (b)

Let AB be the surface of the lake and p be the point of observation such that AP=25 m, Let C be the position of the cloud and and c' be its reflection in the lake. The CB=C'B. Let PM be perpendicular from P on CB. Then, $\angle CPM = 15^0$ and $\angle C'PM = 45^0$. Let CM = h, then CB= h + 25 consequently, CB = h + 25.

In ΔCMP, we have

Tan $15^0 = \dfrac{CM}{PM}$

$\Rightarrow\ 2-\sqrt{3}\ \dfrac{h}{PM}$

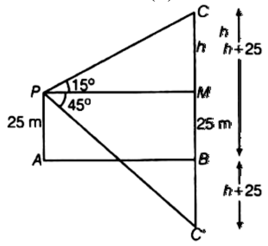
$\Rightarrow PM = \dfrac{h}{2-\sqrt{3}}$ ….(i)

In ΔPMC', we have

$\tan 45^0 = \dfrac{C'M}{PM}$

$\Rightarrow\quad 1 = \dfrac{h+50}{PM}$

$\Rightarrow\quad PM = h+50$ …(ii)



From (i) and (ii)

$\dfrac{h}{2-\sqrt{3}} = h+50$

$\Rightarrow h = h\left(2-\sqrt{3}\right)+50\left(2-\sqrt{3}\right)$

$\Rightarrow h-2h+\sqrt{3h} = 50\left(2-\sqrt{3}\right)$

$\Rightarrow h\left(\sqrt{3}-1\right) = 50\left(2-\sqrt{3}\right)$

$\Rightarrow h\dfrac{50\left(2-\sqrt{3}\right)}{\left(\sqrt{3}-1\right)}$

$\therefore$ Height of the cloud above the lake level= h + 25

$= \dfrac{100-50\sqrt{3}}{\sqrt{3}-1}+25$

$= \dfrac{100-50\sqrt{3}+25\sqrt{3}-25}{\sqrt{3}-1}$

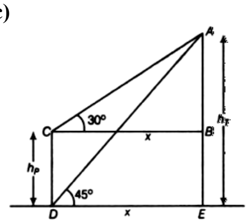
$= \dfrac{75-25\sqrt{3}}{\sqrt{3}-1} = \dfrac{25\left(3-\sqrt{3}\right)}{\sqrt{3}-1}$

$= \dfrac{25\left(3-\sqrt{3}\right)}{\sqrt{3}-1} = \dfrac{\sqrt{3}+1}{\sqrt{3}+1}$

$= \dfrac{25\left(3\sqrt{3}+3-3-\sqrt{3}\right)}{3-1}$

$= \dfrac{25\left(2\sqrt{3}\right)}{2} = 25\sqrt{3}\ m$

Sol.28. (c)



Let CD be the pole and AE be the tower.

Now, in ΔABC,

$\tan 30^0 = \dfrac{AB}{BC}$

$\Rightarrow\ \dfrac{1}{\sqrt{3}} = \dfrac{h_r - h_p}{x}$

$\Rightarrow\ x = \sqrt{3}\left(h_r - h_p\right)$ ...(i)

Now, in ΔAED, $\tan 450^0 = \dfrac{AE}{DE}$

$\Rightarrow\quad 1 = \dfrac{h_r}{x}$

$\Rightarrow\quad x = h_r$ …(ii)

From (i) and (ii), we get $h_r = \sqrt{3}\left(h_r - h_p\right)$

$\Rightarrow h_r = \sqrt{3}h_r - \sqrt{3}h_p$

$\Rightarrow \left(\sqrt{3}-1\right)h_r = \sqrt{3}h_p$

$\Rightarrow\ h_r = \dfrac{\sqrt{3}}{\sqrt{3}-1}h_p$

Now $\dfrac{2h_p}{3+\sqrt{3}} = \dfrac{2}{3+\sqrt{3}}h_p.\dfrac{\sqrt{3}}{\sqrt{3}-1}h_p$

$= \dfrac{2\sqrt{3}}{3\sqrt{3}-3+3-\sqrt{3}}h_p^2$

$= \dfrac{2\sqrt{3}}{2\sqrt{3}}h_p^2 = h_p^2$

Again, $\dfrac{h_r - h_p}{\sqrt{3}+1} = \dfrac{\dfrac{\sqrt{3}}{\sqrt{3}-1}h_p - h_p}{\sqrt{3}+1}$

$= \dfrac{\sqrt{3}h_p - \left(\sqrt{3}-1\right)h_p}{\left(\sqrt{3}-1\right)\left(\sqrt{3}+1\right)}$

$= \dfrac{\sqrt{3}h_p - \sqrt{3}h_p + h_p}{3-1} = \dfrac{h_p}{2}$

Again, $\dfrac{2\left(h_p + h_r\right)}{h_p} = \dfrac{2\left(h_p + \dfrac{\sqrt{3}}{\sqrt{3}-1}h_p\right)}{h_p}$

$= 2\left(\dfrac{\sqrt{3}-1+\sqrt{3}}{\sqrt{3}-1}\right) = 2\left(\dfrac{2\sqrt{3}-1}{\sqrt{3}-1}\right)$

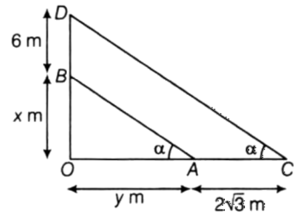
$= \dfrac{4\sqrt{3}-2}{\sqrt{3}-1} = \dfrac{\left(4\sqrt{3}-2\right)\left(\sqrt{3}+1\right)}{\left(\sqrt{3}-1\right)\left(\sqrt{3}+1\right)}$

$= \dfrac{12+4\sqrt{3}-2\sqrt{3}-2}{3-1}$

$= \dfrac{10+2\sqrt{3}}{2} = 5+\sqrt{3}$

Sol.29. (a)

Let OB and BD be the tower and flag-staff respectively. OA and AC be the shadow of tower and flag-staff respectively.



Again let α be the angle that sun makes with the ground.

$\therefore\ \angle OAB = \angle OCD = \alpha$

Now, in ΔOAB

$\tan\alpha = \dfrac{x}{y}$ ….(i)

and in ΔOCD

$\tan\alpha = \dfrac{x+6}{y+2\sqrt{3}}$ …(ii)

from eqs. (i) and (ii) we get $\dfrac{x}{y} = \dfrac{x+6}{y+2\sqrt{3}}$

$\Rightarrow xy + 2\sqrt{3}x = xy + 6x$

$\Rightarrow \dfrac{x}{y} = \sqrt{3}$

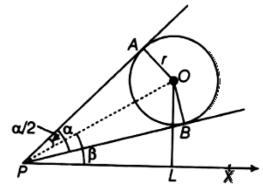
$\Rightarrow \tan\alpha = \sqrt{3}\left[\text{from Eq. }(i)\right]$

$\Rightarrow \alpha = 60^0$

Sol.30. (a)

Let O bet the centre of the balloon, p

Be the eye of the observer and $\angle APB = \alpha$



$\therefore\ \angle APO = \angle BPO = \dfrac{\alpha}{2}$

In ΔOAP

$\sin\dfrac{\alpha}{2} = \dfrac{OA}{OP}$

$\Rightarrow \sin\dfrac{\alpha}{2} = \dfrac{r}{OP} \Rightarrow OP = r\ \text{cosec}\dfrac{\alpha}{z}$

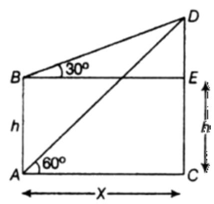
In ΔOPL,

$\sin\beta = \dfrac{OL}{OP}$

$\Rightarrow\ OL = OP\sin\beta$ [from Eqs.(i)]

$\therefore\ OL = \dfrac{r\sin\beta}{\sin\left(\dfrac{\alpha}{2}\right)}$

Sol.31. (b)



In ΔACD

$\tan 60^0 = \frac{H}{x}$

$\Rightarrow \quad \sqrt{3} = \frac{H}{x}$

$\Rightarrow \quad x = \frac{H}{\sqrt{3}}$ ...(i)

In ΔBDE

$\tan 30^0 = \frac{H-h}{x}$

$\Rightarrow \quad \frac{1}{\sqrt{3}} = \frac{H-h}{x}$

$\Rightarrow \quad x = \sqrt{3}(H-h)$

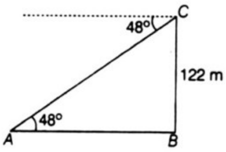
$\Rightarrow \quad H = 3(H-h)$

$\Rightarrow \quad H = 3H - 3h$

$\Rightarrow \quad 2H = 3h$

$\Rightarrow \quad H = \frac{3}{2}h$

Sol.32. (b)



$\tan 48^0 = \frac{BC}{AB}$

$\tan 48^0 = \frac{122}{AB}$

AB= $\frac{122}{\tan 48^0} = 122 \cot 48^0$

$= 122 \cot (90^0 - 42^0)$

$\left[\because \quad -\theta) = \tan\theta\right]$

$= 122 \tan 42^0$

Sol.33. (a)

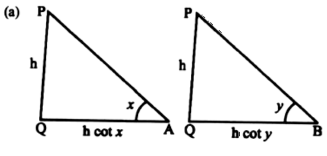
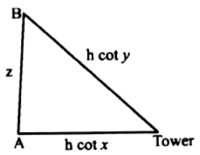


fig. (i) fig. (ii)

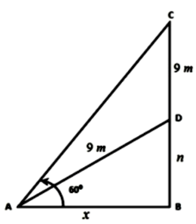


$AQ^2 + AB^2 = BQ^2$

$h^2\cot^2 x + z^2 = h^2\cot^2 y$

$h^2(\cot^2 y - \cot^2 x) = z^2$

Sol.34. (c)



InΔABC

$\tan 60 = \frac{9+n}{x} \quad x = \frac{9+n}{\sqrt{3}}$

In $\Delta ADB \; n^2 + x^2 = 9^2$

$n^2 + \left(\frac{9+n}{\sqrt{3}}\right)^2 = 81$

By solving it n=4.5 m

Height of flagstaff (BC) = 9+4.5 = 13.5 m

Sol.35. (d)

Same diagram of above question

In ΔABC

$\tan 75° = \frac{6+n}{x}$

$2+\sqrt{3} = \frac{6+n}{x}$

$x = \frac{6+n}{2+\sqrt{3}}$

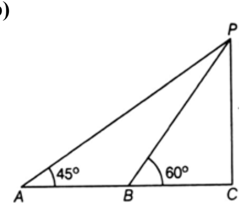
In ΔADB $n^2 + x^2 = 6^2$

$n^2 + \left(\frac{6+n}{2+\sqrt{3}}\right)^2 = 36$

By solving it n = 3√3 cm

Height of flagstaff (BC) = 6 + 3√3 m

Sol.36. (b)



in ΔPBC

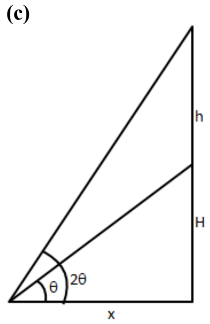
$\tan 60° = \frac{PC}{BC} = \frac{5(3+\sqrt{3})}{x}$

$x = \frac{5(3+\sqrt{3})}{\sqrt{3}} = 5(\sqrt{3}+1)$

in ΔPAC, AB + BC = PC

AB = $5(3+\sqrt{3}) - 5(\sqrt{3}+1) = 10\,\text{m}$

Sol.37. (c)



Let height of tower = H

$\tan 2\theta = \frac{h+H}{x}$ …(1)

$\tan\theta = \frac{H}{x}$ …(2)

divide (2) by (1)

$\frac{\tan 2\theta}{\tan\theta} = \frac{h+H}{h}$

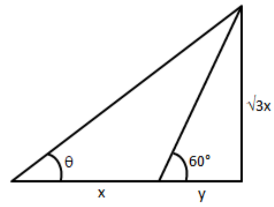
$\frac{2\tan\theta}{\tan\theta(1-\tan^2\theta)} = \frac{h}{H} + 1$

$\frac{2}{1-\tan^2\theta} - 1 = \frac{h}{H}$

$\frac{1+\tan^2\theta}{1-\tan^2\theta} = \frac{h}{H} \Rightarrow \frac{1}{\cos 2\theta}$

$\Rightarrow H = h\cos 2\theta$

Sol.38. (b)



$\tan 60° = \frac{\sqrt{3}x}{y}$

$\sqrt{3} = \frac{\sqrt{3}x}{y}$

x=y

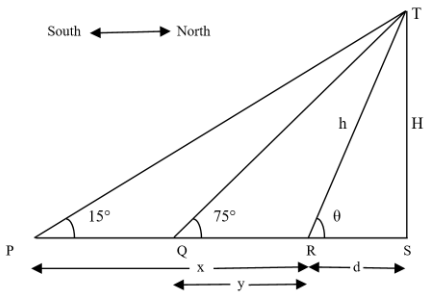
$\tan\theta = \frac{\sqrt{3}x}{x+y} = \frac{\sqrt{3}x}{x+x} = \frac{\sqrt{3}}{2} = 0.866$

$\tan 30° = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3} = 0.577$

tan45° = 1

so 30° < θ < 45°

Sol.39. (a)



In ΔPTS

$\tan 15° = \frac{H}{x+d}$

$x+d = \frac{H}{2-\sqrt{3}} = H(2+\sqrt{3})$ ……….(i)

In ΔQTS

$\tan 75° = \frac{H}{y+d}$

$y+d = \frac{H}{2+\sqrt{3}} = H(2-\sqrt{3})$ ……….(ii)

eq(i) – eq(ii)

$x - y = H(2\sqrt{3})$

$H = \frac{x-y}{2\sqrt{3}}$ ……..(iii)

Sol.40. (d)

In ΔRTS

$\cot\theta = \frac{d}{H}$

Divide eq(i) by eq(ii)

$\frac{x+d}{y+d} = \frac{H(2+\sqrt{3})}{H(2-\sqrt{3})}$

$\frac{x+d}{y+d} = \frac{(2+\sqrt{3})}{(2-\sqrt{3})}$

apply componendo and dividendo

$\frac{x+y+2d}{x-y} = \frac{2}{\sqrt{3}}$

$x+y+2d = \frac{2(x-y)}{\sqrt{3}}$

$$2d=\frac{2(x-y)}{\sqrt{3}}-x-y$$

$$d=\frac{2(x-y)-\sqrt{3}(x+y)}{2\sqrt{3}} \quad \text{……..(iv)}$$

divide (iv) by ((iii)

$$\cot\theta=\frac{d}{H}=\frac{\frac{2(x-y)-\sqrt{3}(x+y)}{2\sqrt{3}}}{\frac{x-y}{2\sqrt{3}}}$$

$$=\frac{\frac{2(x-y)-\sqrt{3}(x+y)}{2\sqrt{3}}}{\frac{x-y}{2\sqrt{3}}}=\frac{2(x-y)-\sqrt{3}(x+y)}{x-y}$$

$$=2-\frac{\sqrt{3}(x+y)}{x-y}$$

Sol.41. (b)

length of tower $h^2=d^2+H^2$

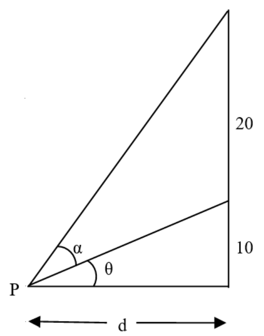
$$h=\sqrt{d^2+H^2}$$

$$h=H\sqrt{1+\left(\frac{d}{H}\right)^2}$$

$$h=H\sqrt{1+\left(2-\frac{\sqrt{3}(x+y)}{x-y}\right)^2}$$

$$h=\left(\frac{x-y}{2\sqrt{3}}\right)\sqrt{1+\left(2-\frac{\sqrt{3}(x+y)}{x-y}\right)^2}$$

Sol.42. (a)



given that

$$\alpha=\tan^{-1}\left(\frac{1}{2}\right)\Rightarrow\tan\alpha=\frac{1}{2} \quad \text{…..(i)}$$

$$\tan\theta=\frac{10}{d} \quad \text{….(ii)}$$

$$\tan(\alpha+\theta)=\frac{30}{d} \quad \text{….(iii)}$$

divide (iii) by (ii)

$$\frac{\tan(\alpha+\theta)}{\tan\theta}=\frac{30}{10}=3$$

$$\tan(\alpha+\theta)=3\tan\theta$$

$$\frac{\tan\alpha+\tan\theta}{1-\tan\alpha\tan\theta}=3\tan\theta$$

$$\frac{\frac{1}{2}+\frac{10}{d}}{1-\frac{1}{2}\cdot\frac{10}{d}}=\frac{30}{d}$$

$$\frac{d+20}{2d-10}=\frac{30}{d}$$

$d^2+20d=60d-300$

$d^2-40d+300=0$

d = 10 or 30

ratio of both values is 1 : 3

statement 1 is correct.

height of flagstaff is 20 that is not equal to 10 and 30.

statement 2 is incorrect.

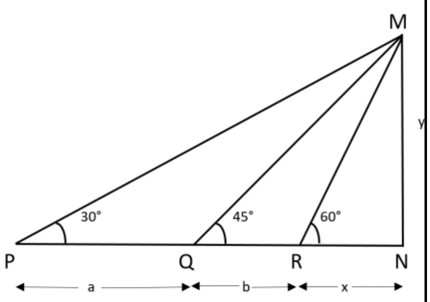
Sol.43. (c)

$$\tan\theta=\frac{10}{d}$$

d = 10 or 30

so $\tan\theta$ = 1 or 1/3

Sol.44. (b)



In ΔMRN

$\tan 60^\circ = y/x$

$y=\sqrt{3}x$ ……..(i)

In ΔMPN

$$\tan 30^\circ=\frac{y}{a+b+x}$$

$a+b+x=\sqrt{3}y$

$a+b+x=3x$

$b=2x-a$ ……..(ii)

In ΔMRNQ

$$\tan 45^\circ=\frac{y}{b+x}$$

$y=b+x$

$\sqrt{3}x=b+x$ …….(iii)

$\sqrt{3}x=2x-a+x$

$a=3x-\sqrt{3}x$

$$x=\frac{a}{3-\sqrt{3}} \quad \text{……..(iv)}$$

PN = a + b + x

= a + 2x – a + x

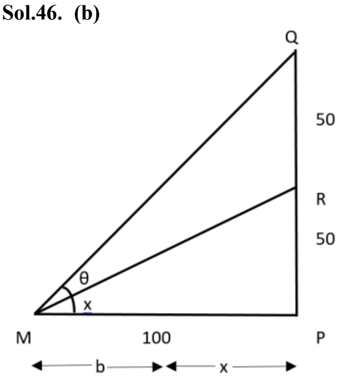
$$=3x=\frac{3a}{3-\sqrt{3}}=\left(\frac{3+\sqrt{3}}{2}\right)a$$

Sol.45. (a)

By eq (iii) in above solution

$$MN=\sqrt{3}x=\sqrt{3}\left(\frac{b}{\sqrt{3}-1}\right)=\left(\frac{3+\sqrt{3}}{2}\right)b$$

Sol.46. (b)



$$\tan x=\frac{50}{100}=\frac{1}{2}$$

$$\tan(\theta+x)=\frac{\tan\theta+\tan x}{1-\tan\theta\tan x}$$

$$\tan(45^\circ)=\frac{\tan\theta+\frac{1}{2}}{1-\tan\theta\frac{1}{2}}=1$$

$$\tan\theta=\frac{1}{3}$$

Solution Videos

Height and Distance 2011 2012 2013

Height and Distance 2014 2015 2016

Height and Distance 2017 2018 2019

Height and Distance 2020 2021 2022

Height and Distance 2023 2024 2025